

Risques climatiques — M2 Actuariat

Climate model: emissions, concentrations and temperature

Gauthier Vermandel

Sorbonne Université — M2 Actuariat

Wednesday, 2 September 2026, 10:45–12:45

Objectives

- Understanding the **basics** of climate models to predict world temperatures.
- Simulating climate models on historical data and make some forecasts conditional on emission scenarios.

Materials:

- `Climate_Models_Notebook.ipynb` *notebook introduction of economic models.*
- `Climate_Models.py` *package to simulate reduced form climate models.*

Introduction

Intro: Why should economists care about climate science?

- Climate change is driven by physical processes that can be **summarized in a simple causal chain**.
- Integrated Assessment Models (IAMs) like DICE or FAIR rely on this chain.
- To interpret or extend IAMs, we need to understand the building blocks.

The Climate-Economy Chain

Emissions → Concentrations → Radiative Forcing → Temperature → Impacts

- Each arrow corresponds to a physical process.
- IAMs implement reduced-form versions of these processes.

Intro: Climate science is about big models

How do climate scientists study the system?

- **General Circulation Models (GCMs)** and **Earth System Models (ESMs)**:
 - Simulate atmosphere, ocean, cryosphere, land surface, and biogeochemical cycles.
 - Built on physical laws (fluid dynamics, thermodynamics, radiative transfer).
 - Used in IPCC reports for future climate projections.
- Strengths:
 - High fidelity, spatial detail, many processes included.
 - Provide benchmarks for scientific understanding.
- Limitations for Integrated Assessment Modelling:
 - Extremely computationally intensive (supercomputers).
 - Results not easily integrated into optimization or policy models.

Today's focus: Economics of climate change

- We will work with **Reduced-form Climate Models (RCMs)**:
 - Tractable and transparent → usable in economic models.
 - Capture the key physical mechanisms without full Earth System complexity.

The Climate-Economy Chain

Emissions → Concentrations → Radiative Forcing → Temperature

Roadmap: We will follow this chain step by step, from *emissions inputs* to *temperature outcomes*, highlighting at each stage how reduced-form models approximate the underlying processes.

Emissions (the inputs)

Emissions: What are emissions?

- **Definition:** Anthropogenic greenhouse-gas (GHG) *flows* into the atmosphere (mainly CO₂, CH₄, N₂O).
- **Focus today:** CO₂ (fossil fuel + land use). Others can be added (e.g., CH₄) as separate inputs.
- Emissions are flows $\Rightarrow$ they feed atmospheric *stocks* in the carbon cycle.

Emissions: Three main green house gases

The three main sources:

- **Carbon dioxide (CO_2)**
 - From fossil fuel combustion, cement, land-use change.
 - Long-lived: persists for centuries.
 - $\sim 76\%$ of global GHG emissions ($\text{CO}_2\text{-eq}$).
- **Methane (CH_4)**
 - From agriculture (rice, livestock), fossil fuel extraction, waste.
 - Short lifetime (~ 10 years) but high warming power.
 - $\sim 16\%$ of global GHG emissions.
- **Nitrous oxide (N_2O)**
 - From fertilizers, biomass burning, industry.
 - Lifetime ~ 120 years.
 - $\sim 6\text{--}7\%$ of global GHG emissions.

To summarize this section, you only need to remember:

- In principle, models can take three distinct exogenous inputs:

$$\{E_t^{CO_2}\}_{t \geq 0}, \quad \{E_t^{CH_4}\}_{t \geq 0}, \quad \{E_t^{N_2O}\}_{t \geq 0}.$$

- In simplified reduced-form models, only CO₂ emissions are modeled explicitly; CH₄ and N₂O are often represented by an additional exogenous forcing term.

Concentrations

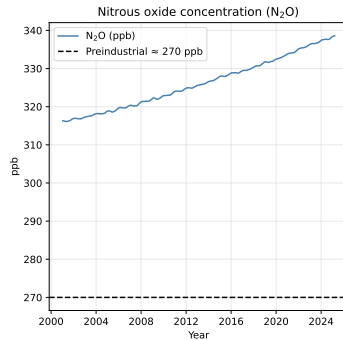
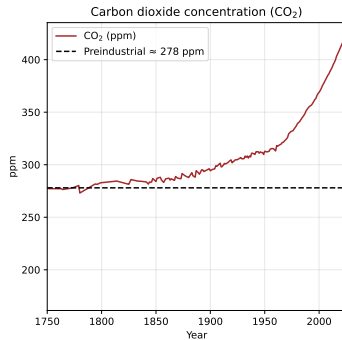
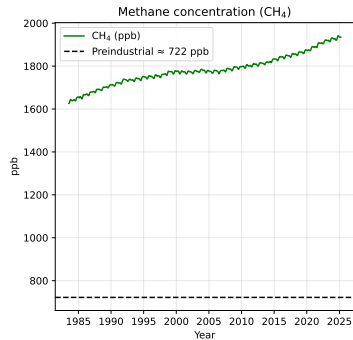
Concentrations: from to emissions to concetration

[Emissions (flows)] $\rightarrow$ [Atmospheric Concentration (stock)] $\rightarrow \dots$

From emissions (flows) to concentrations (stocks):

- Climate does not react directly to annual emissions, but to the **cumulative stock of gases** in the atmosphere.
- Emissions are measured in flows (e.g. GtC/yr).
- Once emitted, gases accumulate in the atmosphere.
- What matters for climate forcing is the concentration (ppm or ppb).

Cnocentrations: Atmospheric concentrations



Concentrations: The mathematics of concentrations

Matrix difference equation (step Δ): t indexes simulation periods and Δ is the length of one period in years.

$$\mathbf{M}_{g,t+1} = \mathbf{M}_{g,t} + \Delta \mathbf{A}_g (\mathbf{M}_{g,t} - \mathbf{M}_{g,0}) + \Delta \mathbf{b}_g E_t^g$$

- $\mathbf{M}_{g,t} \in \mathbb{R}^{n_g}$: vector of compartment stocks/concentrations at t .
- $\mathbf{M}_{g,0}$: baseline (natural equilibrium) vector (e.g. preindustrial levels).
- $\mathbf{A}_g \in \mathbb{R}^{n_g \times n_g}$: *transfer matrix* (diagonal entries < 0 , off-diagonals ≥ 0); row sums ≤ 0 reflect net losses to sinks.
- $\mathbf{b}_g \in \mathbb{R}^{n_g}$: injection vector indicating which box receives anthropogenic emissions.
- E_t^g : emissions flow (units depend on gas).

Modeling choice. The number of boxes n_g determines precision:

- $n_g = 1$: a single decay process (e.g. CH_4 , N_2O).
- $n_g = 2-3$: separates atmosphere, fast reservoirs (biosphere/upper ocean), and slow reservoirs (deep ocean) for CO_2 .
- Larger n_g : more realistic sequestration processes (soil carbon, permafrost, multiple ocean layers) at the cost of extra parameters.

Conservation:

- the system is physical, so total mass is conserved: transfers occur *between* boxes, with no loss outside the system (except for explicitly modelled sinks).

CO₂ : three-box carbon cycle

$$\mathbf{M}_{t+1}^{CO_2} = \Phi^{CO_2} \mathbf{M}_t^{CO_2} + \Delta \Xi^{CO_2} E_t^{CO_2}, \quad \mathbf{M}_t^{CO_2} = \begin{bmatrix} M_{AT} \\ M_{UP} \\ M_{LO} \end{bmatrix} \in \mathbb{R}^3$$

- $\Phi^{CO_2} \in \mathbb{R}^{3 \times 3}$: transfer matrix. Diagonal: persistence (< 1). Off-diagonal: transfers across boxes.
- $\Xi^{CO_2} \in \mathbb{R}^{3 \times 1}$: injection vector (typically only atmosphere receives emissions).
- Boxes: **AT** = atmosphere, **UP** = upper ocean/biosphere, **LO** = deep ocean.

CH_4 : one-box decay to baseline

$$M_{t+1}^{\text{CH}_4} = M_t^{\text{CH}_4} - \Delta \delta_{\text{CH}_4} (M_t^{\text{CH}_4} - M_0^{\text{CH}_4}) + \Delta \kappa_{\text{CH}_4} E_t^{\text{CH}_4}$$

- $M_0^{\text{CH}_4}$: preindustrial baseline (≈ 722 ppb).
- δ_{CH_4} : decay rate ($\tau \sim 12$ y lifetime).
- κ_{CH_4} : conversion emissions $\rightarrow$ ppb/yr ($1 \text{ ppb} \approx 2.78 \text{ MtCH}_4$).

Radiative Forcing

[Emissions (flows)] $\rightarrow$ [Concentration (stock)] $\rightarrow$ [Forcing (flow)] $\rightarrow \dots$

Key idea. Radiative forcing is the *side effect of rising concentrations*: it increases the thickness of the atmospheric “blanket” that traps heat.

- **Baseline:** in the preindustrial climate, incoming solar $\approx$ outgoing infrared (energy balance).
- **Forcing:** any perturbation that shifts this balance.
 - Positive forcing $\Rightarrow$ warming (extra blanket from GHGs).
 - Negative forcing $\Rightarrow$ cooling (reflective aerosols, volcanic dust).
- **Units.** Watts per square meter (W/m^2).

Forcing: General Formula

Key idea. Total forcing is the sum of contributions from different agents, each measured relative to a natural (preindustrial) baseline.

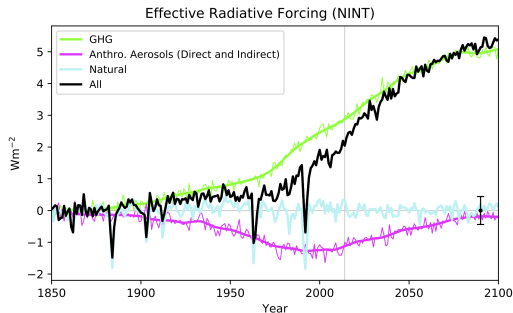
General formulation

$$F_t = \sum_{g \in \{\text{CO}_2, \text{CH}_4, \text{N}_2\text{O}, \text{others}\}} F_t^g$$

$$F_t^g = f_g(M_t^g) - f_g(M_0^g)$$

- M_t^g = atmospheric concentration of gas g at time t .
- M_0^g = preindustrial reference level (e.g. 278 ppm CO₂, 722 ppb CH₄).
- $f_g(\cdot)$ = functional form mapping concentration to forcing.
 - CO₂: logarithmic in concentration.
 - CH₄, N₂O: square-root formulations (with overlap correction).
 - Other agents: aerosols (negative forcing), solar variability, land-use.

Forcing: Some illustration



- **Pink line:** Anthropogenic aerosols $\Rightarrow$ *negative forcing*.
e.g. sulfate aerosols (from burning coal and oil- SO_2 emissions), very reflective: scatter sunlight back to space, short lifetime (days-weeks).
- **Cyan line:** Natural forcings (solar variability, volcanic eruptions).

Forcing: CO₂ Forcing

Forcing formulas map concentrations into energy imbalance (W/m²), measured relative to preindustrial levels.

CO₂ forcing

$$F_t^{CO_2} = F_{2 \times CO_2} \cdot \frac{\ln(M_t^{CO_2} / M_0^{CO_2})}{\ln 2}$$

- Preindustrial $M_0^{CO_2} = 278$ ppm
- Forcing at doubling: $F_{2 \times CO_2} \approx 3.7$ W/m²
- Today ($M_t^{CO_2} \approx 420$ ppm):

$$F_t^{CO_2} \approx 3.7 \cdot \frac{\ln(420/278)}{\ln 2} \approx 2.1 \text{ W/m}^2$$

CH₄ forcing

$$F_t^{CH_4} = \alpha_{CH_4} \left(\sqrt{M_t^{CH_4}} - \sqrt{M_0^{CH_4}} \right)$$

- Preindustrial $M_0^{CH_4} = 722$ ppb
- Forcing coefficient $\alpha_{CH_4} \approx 0.036 \text{ W/m}^2/\sqrt{\text{ppb}}$
- Today ($M_t^{CH_4} \approx 1860$ ppb):

$$F_t^{CH_4} \approx 0.036 \cdot (43.1 - 26.9) \approx 0.6 \text{ W/m}^2$$

Temperatures (output)

[Emissions (flows)] $\rightarrow$ [Concentration (stock)] $\rightarrow$ [Forcing (flow)] $\rightarrow$ [Temperature (stock)]

- **Key point:** Radiative forcing $\neq$ instantaneous warming.

Temperatures behave as a *state variable*: they accumulate the effect of past forcings, with strong inertia.

- **Sources of inertia:**
 - The ocean absorbs most of the excess heat, releasing it slowly.
 - Ice sheets and cryosphere respond on multi-decadal to millennial scales.
 - Feedbacks (clouds, carbon cycle) operate with delays.
- **Illustration:** Even if CO₂ concentrations stabilized today, global temperatures would still rise for decades (“committed warming”).

Temperatures: Warming as an ODE

As with concentrations, temperatures evolve as a linear dynamic system: forcing is a flow, temperatures are stocks.

General formulation

$$\mathbf{T}_{t+1} = \mathbf{T}_t + \Delta(\mathbf{A}_T \mathbf{T}_t + \mathbf{B}_T F_t), \quad \mathbf{T}_0 = \mathbf{0}$$

- $\mathbf{T}_t$ = vector of temperature anomalies (e.g. atmosphere, upper ocean, deep ocean).
- F_t = total radiative forcing at time t (flow).
- $\mathbf{A}_T$ = matrix of heat exchange and feedback parameters.
- $\mathbf{B}_T$ = vector linking forcing to temperature.
- $\mathbf{T}_0 = \mathbf{0}$: preindustrial baseline (no initial warming).

Temperatures: a two box model

A simple reduced-form system tracks two temperature stocks: atmosphere/surface and deep ocean:

$$T_{t+1}^{AT} = T_t^{AT} + \Delta \left[c_1 F_t - c_1 \frac{F_{2 \times CO_2}}{T_{2 \times CO_2}} T_t^{AT} - c_1 c_3 (T_t^{AT} - T_t^{LO}) \right]$$

$$T_{t+1}^{LO} = T_t^{LO} + \Delta \left[c_4 (T_t^{AT} - T_t^{LO}) \right]$$

- T^{AT} : atmospheric/surface temperature anomaly (fast response).
- T^{LO} : deep-ocean temperature anomaly (slow response, high inertia).
- F_t : total radiative forcing at time t .
- Parameters: c_1 (climate response rate), c_3 (surface–ocean coupling), c_4 (deep-ocean heat uptake), $T_{2 \times CO_2}$ (climate sensitivity).
- Initial condition: $T_0^{AT} = T_0^{LO} = 0$ (preindustrial baseline).

Temperatures: asymptotic dynamics

Model in *relative terms* with respect to a doubling CO_2 and hold it fixed (no other forcings).

- **Forcing:**

$$M_{AT}^{CO_2} = 2M_{AT,0}^{CO_2} \Rightarrow F^{CO_2} = F_{2\times CO_2} \approx 3.6 \text{ W/m}^2$$

- **Dynamics (discrete-time 2-box model):**

$$T_{t+1}^{AT} = T_t^{AT} + \Delta \left[c_1 F_{2\times CO_2} - c_1 \frac{F_{2\times CO_2}}{T_{2\times CO_2}} T_t^{AT} - c_1 c_3 (T_t^{AT} - T_t^{LO}) \right]$$

$$T_{t+1}^{LO} = T_t^{LO} + \Delta c_4 (T_t^{AT} - T_t^{LO})$$

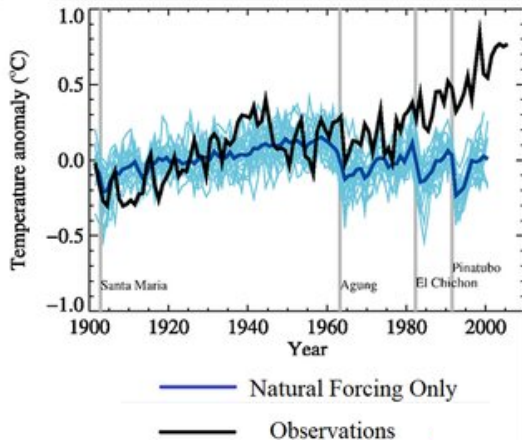
- **Equilibrium:**

$$T^{AT} = T^{LO} = T_{2\times CO_2} \approx 3.1^\circ\text{C}$$

Takeaway: $F_{2\times CO_2}$ = forcing from doubling, $T_{2\times CO_2}$ = long-run warming response.

They anchor reduced-form climate dynamics.

Temperatures: Human vs. Natural Forcings



To wrap up

To summarize the result, consider a carbon pulse of 1GtC:

$+1GtC \rightarrow +0.47 \text{ ppm } CO_2 \rightarrow +0.01 \text{ W/m}^2 \text{ forcing} \rightarrow +0.006^\circ C \text{ in the long term}$

Thank you!

gauthier.vermandel@polytechnique.edu